Hallucination Prediction in Large Language Models Using Contextual Analysis

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Abstract

In recent years, Large Language Models (LLMs) such as ChatGPT have demonstrated impressive capabilities in generation, reasoning, and question answering. However, despite their success, these models frequently hallucinate, producing fluent but factually incorrect outputs with high confidence. This remains a critical barrier to deploying LLMs in high-stakes domains. Existing hallucination detection methods are typically post-hoc, computationally expensive, and lack interpretability. In this work, we introduce a novel, interpretable framework called Contextual Analysis for predicting hallucination in LLMs. Our approach constructs a Hierarchical Knowledge Tree that organizes words based on contextual similarity from the models training data. We propose a new metric, the Word Coverage Score (WCS), which captures both the frequency and contextual depth of prompt terms. Experimental results on a subset of the WikiBio dataset show that prompts with low WCS are significantly more prone to hallucination, validating the predictive utility of our approach. By identifying high-risk prompts before generation, this framework enhances transparency, fosters user trust, and opens new possibilities for more reliable applications of LLMs.

CCS Concepts

• **Computing methodologies** → **Knowledge representation and reasoning**; **Natural language generation**; **Semantic networks**.

Keywords

hallucination prediction, large language models, contextual analysis, explainable AI

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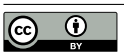
1 Introduction

Large language models (LLMs) such as ChatGPT have taken the world by storm. They have emerged, wielding immense power in their capacity to comprehend linguistic patterns and conjure coherent and contextually appropriate responses. LLMs aim to predict the next token based on previous tokens [27], which primarily concerns their ability to understand and generate basic language. These models are now used for a wide range of natural language processing (NLP) tasks, including machine translation, object recognition, instruction following, and question answering, gaining popularity across various domains such as Medicine, Law, Engineering, Academia, and Finance [10].

However, these models are prone to hallucinations [6, 7, 13], where they generate plausible but factually incorrect or nonsensical information [7]. They produce hallucinations and exhibit biases, yet continue to "speak with confidence" [13]. These systems cannot detect when they provide a false answer, because they are designed to produce text that looks truth-apt without any actual concern for truth [13]. Their quality and accuracy rely on vast amounts of training data that contain inaccuracies and biases, and their internal structures have no inherent mechanism to differentiate truth from falsehood [6].

Recent research has highlighted the need for transparent mechanisms to detect hallucination in LLMs. For instance, TOHA introduces a method that analyses the topology of attention graphs to flag hallucinated content [4], but it lacks deeper mechanistic understanding of the model's internal knowledge structure. Other approaches include activation-based hallucination detection [22] and self-consistent measures [20]. Despite initial successes, these methods remain limited in interpretability; they rarely pinpoint which gaps in LLMs' knowledge drive hallucinations and often operate independently of the prompt structure. Another promising avenue is the knowledge circuit framework, or 'Circuit Transformer,' which identifies sub-networks of attention heads and MLPs underlying factual recall [25]. However, these circuits are typically extracted post-hoc limiting real-time interpretability and require removing or altering specific attention heads, which may not generalize across tasks, models, or layers, making the approach time-consuming and computationally expensive.

Existing studies often fail to link model behavior, particularly hallucinations, to knowledge or contextual data specific to the prompt. Recent evidence shows that low coverage or sparse content within a model's internal representation is strongly correlated with hallucination. For example, Kang et al. demonstrate that LLMs exhibit co-occurrence bias, when subject–object pairs appear infrequently in pretraining data, models struggle to recall them correctly, often producing plausible but incorrect outputs [16]. Similarly, ComparisonQA benchmarks reveal that factual accuracy drops significantly



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on low-frequency facts, even when questions are identical, signifying that sparse exposure directly impairs recall [28]. In other words, rare entities or facts not robustly encoded during training increase the likelihood of hallucination.

Inspired by these insights, we propose an interpretable framework that predicts hallucination risk by analysing prompt content rather than detecting and analysing errors post-generation. We introduce Contextual Analysis (CA), which employs a Hierarchical Knowledge Tree (HKT) to represent word relationships based on shared source contexts, providing a structured, interpretable view of knowledge domains [3]. As illustrated in Figure 1, CA analyses the contextual structure of a prompt by examining how each token is supported contextually; the HKT then predicts areas of low contextual support, which may correlate with hallucination risk. This framework identifies hallucinated content without external fact-checking [24] and offers a transparent method for assessing response validity. Previously, CA has been used for sentiment-analysis performance prediction and short-text topic detection, demonstrating its potential to detect subtle contextual shifts and model degradation [2, 3]. CA's transparency makes it a compelling approach for hallucination prediction, where interpretability is paramount.

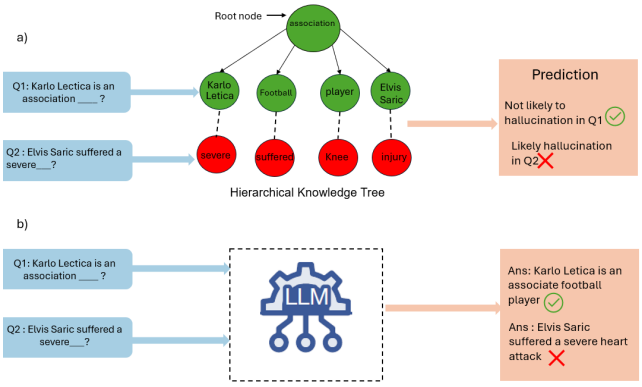


Figure 1: (a) Illustrates the proposed Contextual Analysis workflow. Given an LLM prompt (left), and a previously built Hierarchical Knowledge Tree (HKT), we perform Contextual Analysis in order to predict potential hallucination. The rightmost element in (a) shows hallucination prediction output of the HKT module with respect to these prompts (b) Conventional LLM I/O flow, whose behavior is correctly predicted using the proposed Contextual Analysis. The detailed hallucination prediction mechanism via HKT is explained in Section 3.3

2 RELATED WORK

Numerous methods have been explored to enhance the accuracy and robustness of large language models (LLMs), including the use of more diverse and representative training data, memory augmentation mechanisms, and structured learning strategies. These robustification techniques generally fall into the following categories:

(1) Fine-Tuning and Instruction Tuning, (2) Retrieval-Augmented Generation (RAG), (3) Alignment with Human Feedback, (4) Chain-of-Thought Prompting, and (5) Custom guardrail systems. Each approach targets different sources of hallucination and has been adopted in various applications to increase factual consistency and contextual grounding.

- Fine-Tuning and Instruction Tuning: Fine-tuning on domain-specific or fact-rich datasets can enhance the model's accuracy in constrained environments. Instruction tuning has shown strong promise in reducing hallucinations by guiding LLMs to follow structured prompts [21]. Models like FLAN-T5 and Alpaca leverage task-specific datasets to align generation with intended outputs.
- Retrieval-Augmented Generation (RAG): RAG [24] frameworks LLMs with search engines or document retrieval systems to provide context before or during generation. This significantly reduces hallucination in knowledge-heavy tasks like question answering and summarization. Notable examples include RAG for knowledge-intensive generation [19], code generation [11], and paraphrasing [17] which demonstrate that access to external documents can anchor generation in verifiable facts.
- Alignment with Human Feedback (RLHF): Reinforcement Learning from Human Feedback (RLHF) has emerged as a powerful tool to steer LLMs toward truthful, safe, and coherent outputs. By learning from human-preferred responses, models such as InstructGPT [21]. and GPT-4o [14], significantly outperform base models in hallucination benchmarks.
- Chain-of-Thought Prompting: Chain-of-thought (CoT) prompting encourages an LLM to produce a step-by-step reasoning path before giving a final answer [18]. By making the model's internal reasoning explicit, CoT helps catch inconsistencies and prevents the model from leaping to incorrect conclusions, thereby improving factual accuracy [18]. However, CoT alone does not fully address hallucinations. Subsequent work such as [26] demonstrated that combining CoT with document retrieval significantly reduces hallucinations by anchoring the model's intermediate steps to verifiable sources.
- Custom guardrail systems : Guardrail systems provide an external layer of control that monitors and shapes LLM outputs to avoid factual errors and other unwanted content [15]. Guardrail can be thought of as a safety filter or validation mechanism applied before, during, or after the model generates a response [15]. Unlike pure prompting methods, guardrails may use additional models, rules, or knowledge sources to intercept hallucinations.

Despite several research efforts, current hallucination detection methods for large language models remain limited in transparency and interpretability [8]. Many approaches rely on the same model that produced the hallucination [20] or depend on external verification techniques that require complex retrieval pipelines. While these methods offer incremental improvements, they often suffer from high computational costs and poor explainability, which poses a significant challenge in real-world, high-stakes scenarios where reliability is critical.

In contrast, our proposed method introduces a lightweight and transparent framework based on Contextual Analysis (CA). It offers a structurally interpretable approach for hallucination prediction by leveraging a Hierarchical Knowledge Tree (HKT) to evaluate contextual support for model queries. This method provides a clear path toward trustworthy, explainable, and resource-efficient hallucination prediction, especially in settings where fact-checking is not feasible.

3 METHODOLOGY

In this section, the methodology framework used to detect hallucination in large language models is presented. The research hypothesis, experimental environment including dataset characteristics, and setup is explained in detail, followed by the development and implementation of an evaluation metric.

3.1 Experiment Environment

The experiments were conducted on a laptop equipped with an Intel® Core™ i5 processor, 16GB of RAM, and running on Windows 10 Enterprise. All codes used for this experiment were developed using the following programming languages: C# (for CA), Python (for LLM), and SQL (for metric development).

3.2 Experiment Detail

The experiments were conducted using the WikiBio dataset on individuals who lived in the Republic of Venice, which is available at [9]. Rather than beginning with a large-scale model and dataset, a subset was selected from the WikiBio dataset, consisting of 1,600 biographies. To enable a focused analysis, we selected a representative subset of 13 biographies featuring individuals associated with sports, guided by the principles of Retrieval-Augmented Generation (RAG) systems [5]. RAG systems inherently prioritize semantic coherence and relevance over data volume through embedding-based retrieval methods, where contextually focused document sets enhance retrieval accuracy and mitigate noise [19]. Thus, our choice of a focused subset aligns with established RAG methodologies, allowing for precise and insightful evaluation of the model's behavior without the complexity often introduced by massive dataset corpora. The Biography Generator model used in this experiment is trained on the full WikiBio dataset. It combines the strengths of GPT-2 and XLNet architectures, leveraging both autoregressive and bidirectional learning capabilities. The same 13 biographies subset primarily focused on individuals in sports was selected from the larger WikiBio dataset to serve as input for model queries. This same subset was also used to construct a Hierarchical Knowledge Tree (HKT), designed to capture and visualize contextual relationships within the data. The HKT was built using Contextual Analysis techniques. The first step involved preprocessing each biography: non-alphabetic and non-numeric characters were removed, text was converted to lowercase, tokenized by whitespace between any set of characters, and stop words were removed. This preprocessing was performed using the Microsoft ML library. Following this, the Hierarchical Knowledge Tree (HKT) was constructed using the Contextual Analysis (CA) approach. CA employs the concept of hierarchical knowledge trees to capture relationships between words based on their co-occurrence in shared contexts. This is initiated by creating a lookup table that assigns each word a unique numeric value. The construction of the HKT begins by grouping words into nodes based on their association with the source documents. The process starts with the word that appears in the highest number of sources, referred to as Word1 which forms the initial node. This node is placed inside a container known as the "Seed HKT". Within this container, all additional words must meet a minimum source overlap requirement, governed by a user-defined threshold relative to the source count of Word1. A second user-defined threshold is applied to group words with similar source patterns in the same node. The HKT is built recursively: once a node is formed, the algorithm identifies new words to form sub-nodes, which are then linked hierarchically as children to their respective parent nodes. This recursive process continues until all words and their associated sources are accounted for in the hierarchical structure, allowing for a comprehensive knowledge representation. (For more details about this approach, the reader is referred to the original work [2, 3]).

3.3 Hallucination Prediction Mechanism

To examine the potential hallucination prediction capabilities of Contextual Analysis (CA), a Hierarchical Knowledge Tree (HKT) was implemented in C#. This algorithm constructs a hierarchy of word clusters based on shared sources. Each node in the tree contains a group of word objects with significant overlap in the source they appear in. The root node contains the most frequent words, while subsequent levels contain words that are increasingly rare and contextually specific.The HKT algorithm described above is inspired by the distributional hypothesis, which states that words occurring in a similar context tend to have a similar meaning [23]. This contextual alignment is also closely related to the RAG principles, which similarly emphasize semantic coherence and contextual relevance through embedding-based retrieval [19]. An illustrative example of the HKT structure is presented in Figure 2, derived from the sports-related entity in the dataset. This figure also shows the number of sources in which these words exist, emphasising that frequent words existing in multiple sources are at the root node, which reflects how contextually relevant these words are. The structure of the HKT demonstrates how the number of sources for words at lower levels reduces as we traverse down the tree, showing how the contextual grounding of the words reduces with depth. To evaluate model behaviour, a list of 140 prompts was developed, each built around specific words or entities from the subset dataset. These prompts were submitted to the biography generator model trained on the full WikiBio dataset. The model's response was evaluated against known facts to categorize each outcome into one of the three groups:

- Correct (factually accurate responses)
- Inconsistent (responses containing unreliable & unstable information)
- Hallucinated (factually incorrect or fabricated responses)

Some example prompts and the corresponding model responses are shown in Table 1 below.

Building on the already developed HKT structure, a metric known as Word Coverage Score (WCS) is introduced to capture the contextual relevance of individual words within the prompts, as

Table 1: Example Prompts with LLM Response

No.	Example Prompt	LLM Response	Accuracy
1	Karlo Letica is an association _____?	Karlo Letica is an association Football player	Correct
2	Karlo Letica is a football player who plays as ___?	Karlo Letica is a football player who plays as a defensive midfielder/a goalkeeper/ an offensive midfielder/ an attacking midfielder	Inconsistent
3	Karlo Letica moved abroad to ___?	Karlo Letica moved abroad to vienna	Hallucinated
4	Elvis Sarić suffered a severe ___?	Elvis Sarić suffered a severe heart attack	Hallucinated

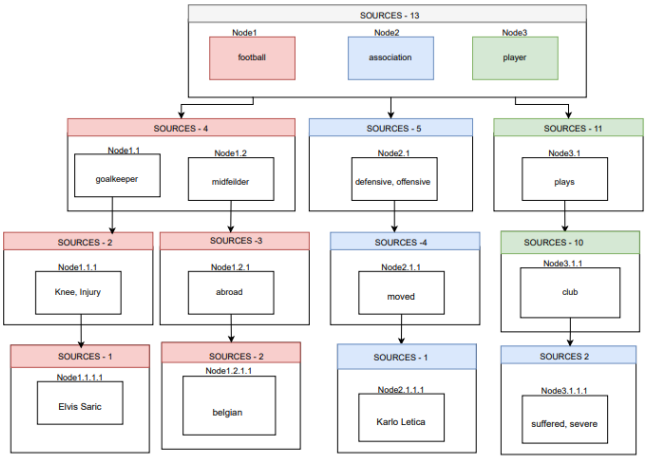


Figure 2: HKT Representation Using CA Algorithm and Dataset

listed in Table 1. This metric capture two key properties for each word in a prompt:

- Coverage: the number of unique source documents in which the word appears.
- Depth: the level of the word's node in the Hierarchical Knowledge tree, where the lower levels represent lower coverage words that are sparse in the source document.

The Word Coverage Score (WCS) for a word b is defined in eq (1) below:

$$WCS(b) = \frac{sources(b)}{level(b)^2 + 1} \tag{1}$$

where :

- $sources(b)$: the number of unique source documents containing word b,
- $level(b)$: the depth of the word's node in the HKT (Level 0 being the root)

The squaring of the level emphasises deep nodes that are sparse in the document containing low coverage words which could trigger hallucination in LLM. The average WCS for each prompt is then calculated by computing the WCS for every word in the prompt and taking the mean.

To illustrate how the WCS is implemented, consider Prompt 1 from Table 1. "Elvis Saric is an association?", this prompt is first tokenized removing stop words like "is", "an" so that the remaining tokens are "Elvis", "Saric", "Association". Using the HKT we determine the source count and level for each word and then calculate the WCS to derive an average score of 3.6852 for the prompt. Furthermore for Prompt 5: "Elvis Saric was diagnosed with?" , the resulting average score is 0.0153 which is significantly lower. These scores are a reflection of how contextually relevant certain words in the prompt are which gives us the predictive capability of detecting factual error by an LLM. Table 2 presents the computed WCS scores using equation 1 and LLM responses for a number of prompts, where it can be observed that lower WCS values are associated with hallucination.

3.4 Evaluation Methodology

In this study a one-way ANOVA was applied with significance level of 0.05 α to identify which group is different from all the three studied groups (correct, inconsistent, and hallucinated) with respect to the proposed metric, then to identify which specific group differences were statistically significant, Tukey's Honest Significant Difference (HSD) test was applied. Both one-way analysis of variance (ANOVA) and the Tukey's HSD test were performed simultaneously by using Python. The interval plot of the WCS score vs group (with 95% CI of mean) from the one-way ANOVA [12] and Tukey's HSD analysis [1] with significance level (95%) are plotted in Figures 3 and 4 respectively . The details of the ANOVA summary and HSD test is presented in Tables 3 and 4.

One-way ANOVA test: The one-way ANOVA test for the scores derived from the proposed metric Word Coverage Score (WCS) was performed to identify whether the specific groups were significantly different from each other or not. The following method was adopted to perform the ANOVA study:

- Null Hypothesis: All means are equal
- Alternative hypothesis: at least one mean is different
- Significance level: $\alpha = 0.05$

The result of the one-way ANOVA of the WCS score vs group revealed that the P value was found to be 2.4×10^{-9} signifying that there is a significant difference that exists in the scores obtained in the different groups. The one-way ANOVA plot in Figure 3 also indicates that the confidence intervals for the group means are clearly separated, suggesting significant differences between them.

Table 2: Comparison of LLM-predicted and actual responses to example prompts

Example Prompt	WCS Score	Predicted LLM output	Actual LLM output
Karlo Letica is an association _____?	3.680	Karlo Letica is an association football player	Karlo Letica is an association football player
Karlo letica is a football player who plays as _____?	0.712	Karlo Letica is a football player who plays as a defensive midfielder	Karlo Letica is a football player who plays as a goal keeper
Karlo Letica is a football player who plays as _____?	0.712	Karlo Letica is a football player who plays as a goal keeper	Karlo Letica is a football player who plays a goal keeper
Karlo Letica moved abroad to_____ ?	0.573	Karlo Letica moved abroad to vienna	Karlo Letica moved abroad to a Belgian Club
Elvis Sarić suffered a severe_____?	0.015	Elvis Sarić suffered a severe heart attack	Elvis Sarić suffered a sever knee injury

Note: Prompts 2 and 3 are intentionally repeated with identical wording. Since the tokens are the same, they yield the same WCS score (0.712). However, the LLM produced different outputs across trials (correct and hallucinated). This is classified as an "inconsistent" behavior of the model as illustrated in Table 1.

Table 3: Analysis of Variance Table for WCS Scores

Source of Variation	SS	df	MS	F	P-value	F-crit
Between Groups	95.29	2	47.64	23.05	2.36E-09	3.06
Within Groups	283.20	137	2.08			
Total	378.49	139				

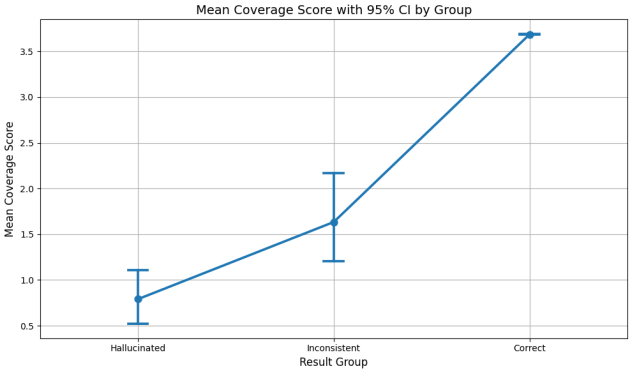


Figure 3: One-way ANOVA plot with 95% CI for the mean

Tukey's HSD Post-Hoc Analysis: In order to identify which specific groups differ significantly from each other, the Tukey's HSD test is applied. The results from Table 4 and Figure 4 below confirmed the following: the Tukey post hoc test revealed significant differences between all group pairs. The adjusted P value for the correct and hallucinated prompts, the correct and inconsistent prompts, and the hallucinated and inconsistent prompts were 0.0, 0.0001 and 0.0063 respectively, which is less than the selected significance level. Therefore, the adjusted p-values for these comparisons also indicate statistically significant differences between all groups.

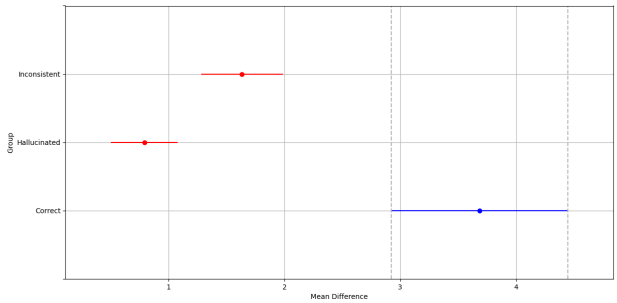


Figure 4: Tukey's HSD plot for WCS Scores

Table 4: Tukey's HSD test for WCS Scores

Group1	Group 2	meandiff	p-adj	lower	upper
Correct	Hallucinated	-2.90	0.0	-3.95	-1.85
Correct	Inconsistent	-2.05	0.0001	-3.17	-0.94
Hallucinated	Inconsistent	0.84	0.0063	0.20	1.48

Evaluation of Prompts using WCS metric: In this section, we evaluate the CA's ability to predict hallucination using the metric described in eq.1. We assess 140 prompts fed into the Biography Generation model. Based on the generated responses, each prompt was labeled into one of three categories: *Correct*, *Inconsistent*, or *Hallucinated*, according to known facts.

To enable prediction of factuality based on WCS, The score distribution was segmented using quartile thresholds derived from the 25th (Q1) and 75th (Q3) percentiles of the WCS values.:

- **Correct**: WCS > 1.3508
- **Inconsistent**: $0.1135 <$ WCS ≤ 1.3508
- **Hallucinated**: WCS ≤ 0.1135

The WCS-based classifications were then compared with actual known facts to evaluate the predictive power of the metric and identify misclassifications.

Table 5: Comparison of WCS-Based Predictions and Actual Known Facts

Known Fact	WCS Prediction			Total
	Correct	Hallucinated	Inconsistent	
Correct	12	0	0	12
Hallucinated	11	32	44	87
Inconsistent	12	3	26	41
Total	35	35	70	140

As shown in Table 5, all prompts labeled as correct were accurately predicted by the WCS-based metric, with no error in prediction. However, the metric mispredicted correctness and inconsistency in the hallucinated and inconsistent cases. Specifically, 11 hallucinated prompts were misclassified as correct, and 44 as inconsistent. Additionally, 12 inconsistent prompts were misclassified as correct and 3 as hallucinated. Overall, the WCS-based metric achieved an accuracy of 50% when compared to the ground truth. If hallucinated and inconsistent responses are included together and as a single group (i.e. these outputs are not to be trusted), then the accuracy increases to 75%.

4 Discussion

The findings of this study demonstrate the effectiveness of the proposed Contextual Analysis (CA) framework in predicting hallucination in large language models (LLMs). By constructing a Hierarchical Knowledge Tree that maps which components of a prompt are contextually relevant and which are not, CA can provide an early warning of potential hallucination prior to output generation, moving beyond post-hoc, output-based detection.

Although LLMs generate fluent and coherent text, their internal representations remain a 'black box'. Despite various efforts to investigate the inner workings of transformers,such as the circuit transformer framework [25], which identifies subnetworks of attention heads and MLPs responsible for specific behaviors, these methods are typically post hoc, computationally intensive, and do not connect model behavior to the knowledge data relevant to a given prompt.

A major strength of the proposed CA framework lies in its explainability and interpretability. Unlike post hoc hallucination detection systems or black-box uncertainty estimators, CA constructs a Hierarchical Knowledge Tree (HKT) that provides a structured and comprehensible representation of the word relationships that exist within the LLM data. By using shared source contexts as the basis for node creation, the HKT allows us to identify which prompts are well-grounded and which are not, based on word frequency and knowledge level. This facilitates the pregeneration risk assessment of hallucination, a significant distinction from methods that rely on analysing the output after it has been generated.

To quantify the predictive utility of this approach, a Word Coverage Score (WCS) metric was derived. When evaluated across 140 prompts, the WCS metric achieved an overall accuracy of 50%. However when hallucinated and inconsistent results are combined together, this rises to 75%. While this result suggests that the CA framework captures meaningful distinctions in contextual grounding, it also highlights the need for refinement. Future work will focus on developing a more nuanced and robust metric to better capture contextual relationship between words and improve hallucination prediction.

In summary, the contextual analysis framework represents a meaningful step forward in transparent and predictive interpretability of language model behaviour. It captures contextual gaps in an intuitive and resource-efficient manner.The introduction of the Word Coverage Score (WCS) provides an initial quantitative basis for this predictive assessment, with promising accuracy, particularly when both hallucinated and inconsistent outputs are considered. However, the current metric still leaves room for improvement, While further work is needed to scale and generalize CA across broader datasets and model types, its foundational design is well positioned to support the growing need for trustworthy and explainable AI in high-stakes applications.

5 Conclusion

This work presents an interpretable and lightweight framework, "Contextual Analysis (CA)" for predicting hallucination in Large Language Models (LLMs). By leveraging a Hierarchical Knowledge Tree (HKT) to model contextual relationships between words in a data source and introducing the Word Coverage Score (WCS) as a novel metric, the approach enables transparent and statistical assessment of hallucination risk. An initial evaluation of WCS on a sample of 140 prompts demonstrated a 50% accuracy in predicting factuality labels, or 75% when hallucinated and inconsistent results are combined together. This gives strong evidence that the metric captures meaningful aspects of contextual grounding while also indicating room for improvement. To ensure a focused evaluation, the initial study was conducted on a subset of the WikiBio dataset. This choice supports efficient experimentation and refinement of the methodology. Ongoing work will expand the scope to include significantly larger and more diverse portions of the WikiBio dataset and newer datasets; this will help to further validate the findings of the proposed method under more complex and diverse data conditions. Ultimately, the proposed framework lays an important foundation for scalable and trustworthy LLM.

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